Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: KMA Dairy

Physical address of dairy:

3737 Weir RD	Livingston	Merced	95334
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X049-X400-X005-XXXX

B. OPERATORS

Azevedo, Kevin

Operator name: Azevedo, Kevin

Telephone no.: (209) 000-0000 (Landline) (Cellular)

3737 Weir RD	Livingston	CA	95334
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Azevedo, Luis

Legal owner name: Azevedo, Luis

Telephone no.: (Landline) (123) 000-0000 (Cellular)

3737 Weir RD	Livingston	CA	95334
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	40	0	0	0
Number under roof	280	20	0	0	0	0
Maximum number	280	20	40	0	0	0
Average number	280	20	40	0	0	0
Avg live weight (lbs)	1,000	1,100	900	0		

Predominant milk cow breed: Holstein

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 7,686.22 *tons per reporting period*

Total nitrogen from manure: 99,204.18 *lbs per reporting period*

After ammonia losses (30% loss applied): 69,442.93 *lbs per reporting period*

Total phosphorus from manure: 16,626.99 *lbs per reporting period*

Total potassium from manure: 52,702.78 *lbs per reporting period*

Total salt from manure: 136,437.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 1,464,000 *gallons*

Total nitrogen generated: 6,128.77 *lbs*

Total phosphorus generated: 889.62 *lbs*

Total potassium generated: 8,655.10 *lbs*

Total salt generated: 43,721.12 *lbs*

	312,000	*gallons applied*
+	1,152,000	*gallons exported*
-	0	*gallons imported*
=	1,464,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal Water	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
12/31/2023	Process wastewater	1,152,000.00 *gal*	452.00	290.00	0.00	0.00	67.00	672.00		3,600

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	4,345.27	644.10	6,460.23	34,608.38
Total exports for all materials	4,345.27	644.10	6,460.23	34,608.38

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
5	7	7	1	process wastewater	X049-X400-X005-XXXX
Totals for areas that were used for application	7	7	1		
Totals for areas that were not used for application					
Land application area totals	7	7	1		

B. CROPS AND HARVESTS

5

Field name: 5

12/22/2022: Oats, hay

Crop: Oats, hay Acres planted: 7 Plant date: 12/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/07/2023	390.00 *ton*	Dry-weight		3.7	14,300.00	2,080.00	29,800.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	55.71	1,534.47	223.20	3,197.71	12,554.77

NUTRIENT BUDGET

A. LAND APPLICATIONS

5 - 12/22/2022: Oats, hay

Field name: 5

Crop: Oats, hay

Plant date: 12/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	137.19	18.89	168.84	700.98	168,000.00 *gal*
Canal Water	Surface water	1.58	0.00	0.00	143.27	1,413,818.18 *gal*
Application event totals		138.78	18.89	168.84	844.25	

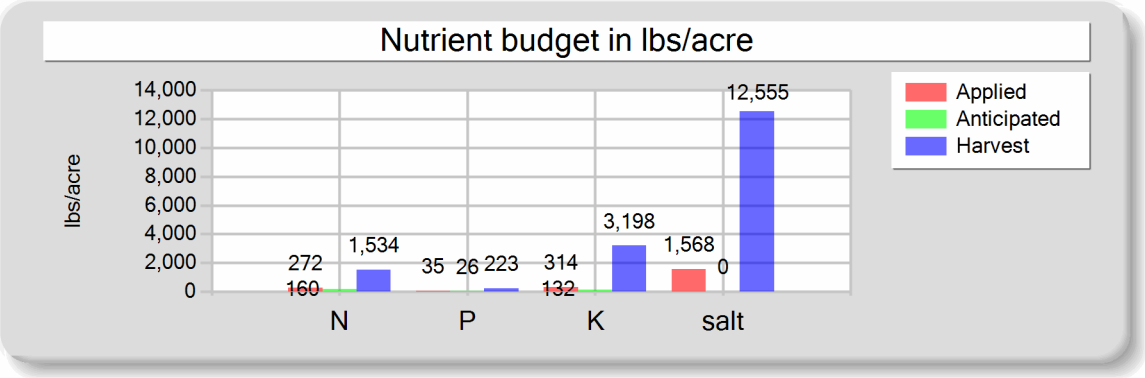
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	117.59	16.19	144.72	600.84	144,000.00 *gal*
Canal Water	Surface water	1.36	0.00	0.00	122.80	1,211,844.15 *gal*
Application event totals		118.95	16.19	144.72	723.64	

B. NUTRIENT BUDGET

5 - 12/22/2022: Oats, hay

Field name: 5 Crop: Oats, hay Plant date: 12/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	254.78	35.07	313.55	1,301.82
Fresh water	2.94	0.00	0.00	266.06
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	271.73	35.07	313.55	1,567.88
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	1,534.47	223.20	3,197.71	12,554.77
Nutrient balance	-1,262.74	-188.12	-2,884.16	-10,986.88
Applied to removed ratio	0.18	0.16	0.10	0.12

Fresh water applied
2,625,662.33 *gallons*
96.69 *acre-inches*
13.81 *inches/acre*

Process wastewater applied
312,000.00 *gallons*
11.49 *acre-inches*
1.64 *inches/acre*

Total harvests for the crop
1 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2308016-01

Sample and source description: Manure 2308016-01

Sample date: 08/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	9,940.00	64,100.00							39.10
DL	5.00	10.00	10.00							0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302056-01

Sample and source description: Lagoon Water 2302056-01

Sample date: 01/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	775.00	775.00			72.10	860.00								6,350.00	4,000
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303394-01

Sample and source description: Lagoon Water 2303394-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	685.00	685.00			94.30	843.00								6,150.00	3,500
DL	0.60	10.00			0.10	10.00								1.00	1

Lagoon Water 2310013-01

Sample and source description: Lagoon Water 2310013-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	452.00	290.00			67.00	672.00								5,210.00	3,600
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Canal Water

2310012-01

Sample description: 2310012-01

Sample date: 09/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.94										28.30	85
DL	0.40										1.00	1

Domestic Well

2310492-01

Sample description: 2310492-01

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											512.00	
DL											1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

5 - 12/22/2022: Oats, hay

2306322-01

Sample and source description: 2306322-01

Sample date: 06/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 3.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	2,080.00	29,800.00		11.70
DL	1.00	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

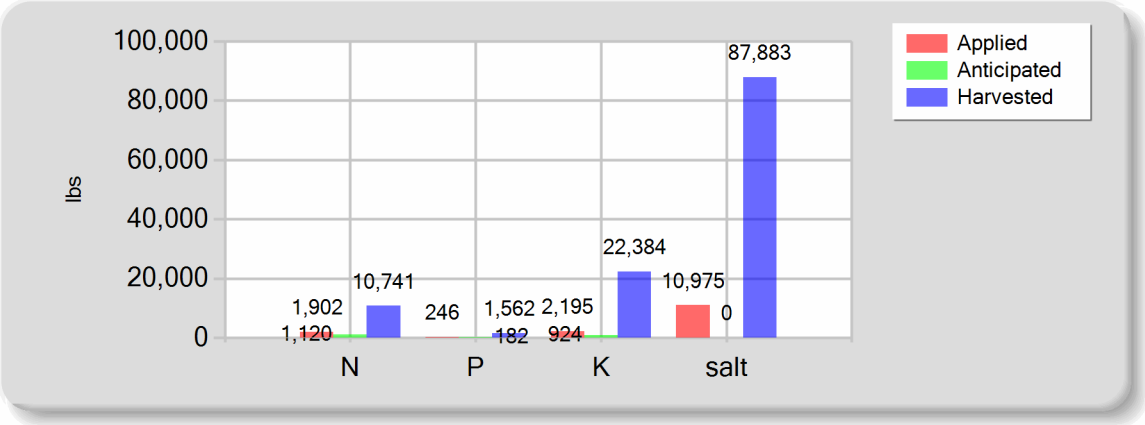
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

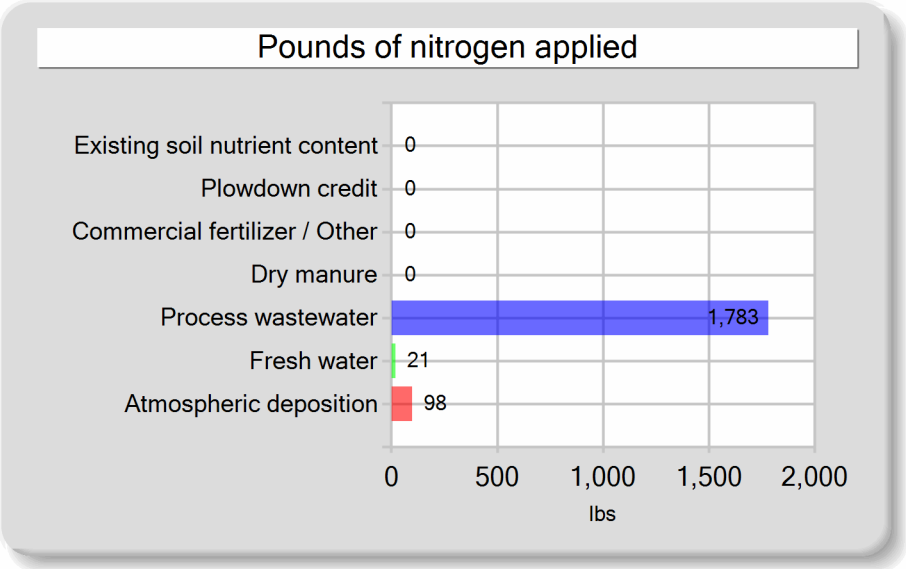
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

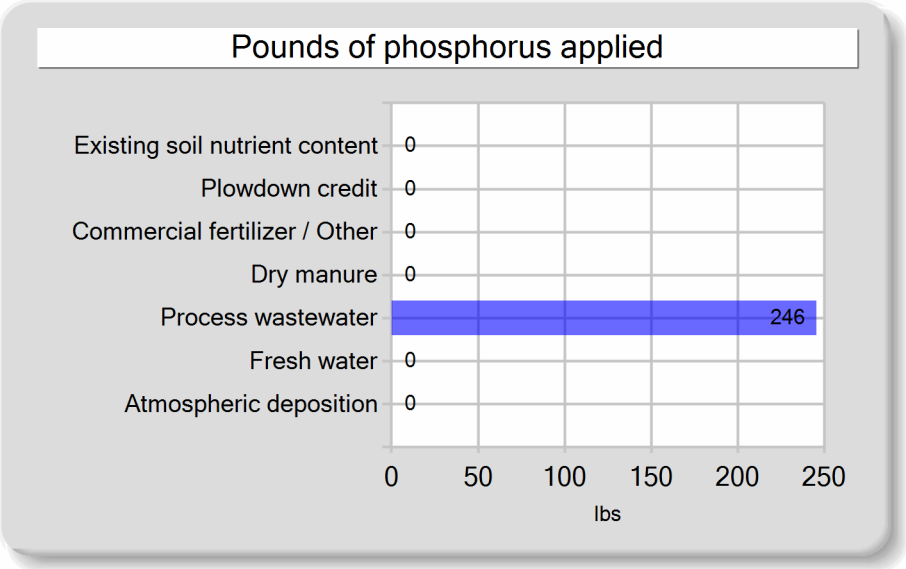
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	1,783.49	245.52	2,194.87	9,112.74
Fresh water	20.60	0.00	0.00	1,862.45
Atmospheric deposition	98.00	0.00	0.00	0.00
Total nutrients applied	1,902.09	245.52	2,194.87	10,975.19
Anticipated crop nutrient removal	1,120.00	182.00	924.00	0.00
Actual crop nutrient removal	10,741.30	1,562.37	22,383.97	87,883.38
Nutrient balance	-8,839.21	-1,316.85	-20,189.10	-76,908.19
Applied to removed ratio	0.18	0.16	0.10	0.12

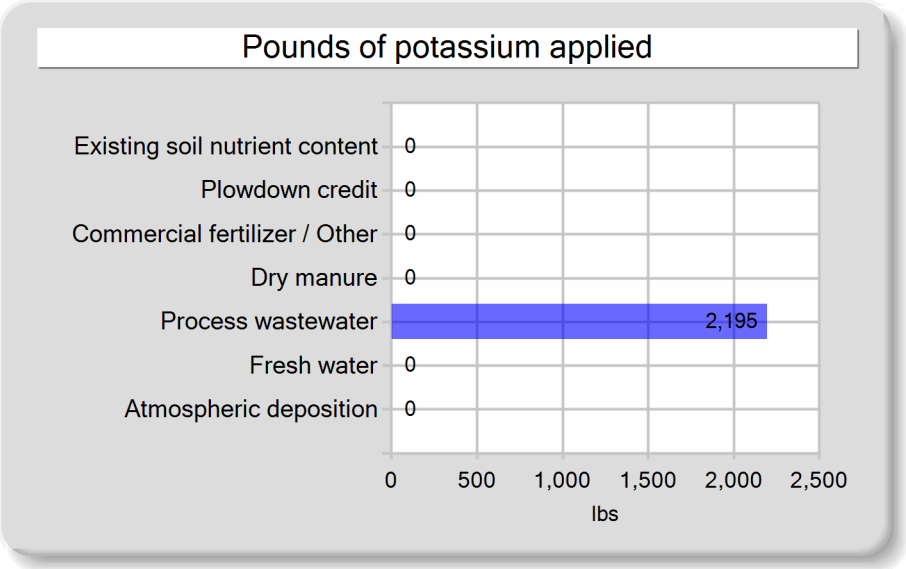
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

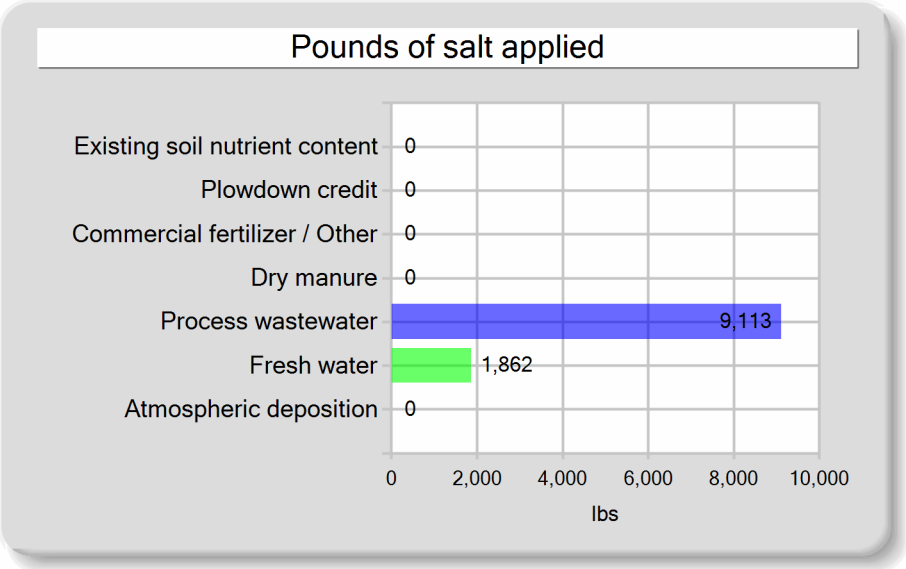


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

*Field #5 was fallow for the spring in 2023.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Luis Azevedo	Kevin Azevedo
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/28/24	6/28/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	See Attachment
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	